

Lab Report Questions for

Research methodology in Computer Applications

Instruction of lab report

General Instructions

1. Statistical data analysis may be performed using **SPSS, Stata, Python, R**, or other relevant statistical software/tools.
2. Research reports and proposals may be prepared using **LaTeX** or other suitable document-processing applications.
3. **Mendeley or Zotero** should be used for literature organization, citation, and referencing.
4. **Turnitin** or an institution-approved plagiarism-detection system should be used to check the originality of the research report.
5. For each practical task, students should submit:
 - Problem statement/question
 - Methodology/procedure
 - Input/data/source
 - Analysis/design
 - Output/results
 - Interpretation
 - Conclusion
 - Screenshots where appropriate

Section A: Questionnaire Design for Data Collection

Question 1: Design a Research Questionnaire

Select a research topic related to **Computer Applications** and design a structured questionnaire for primary data collection.

The questionnaire should contain:

- Title of the research
- Brief introduction/instructions to respondents
- Demographic questions
- Appropriate research variables
- Multiple-choice questions
- Likert-scale questions
- Yes/No questions
- Rating-scale questions
- At least one open-ended question
- Questions mapped to the research objectives

Suggested Computer Application topics:

- Student Satisfaction with Learning Management Systems
- Factors Affecting Adoption of Mobile Applications
- User Satisfaction with E-Government Applications
- Cloud Storage Usage among IT Students
- Use of AI Tools among Computer Application Students
- Online Learning Application Usage
- Cybersecurity Awareness among University Students
- User Experience of Mobile Banking Applications
- Factors Affecting E-Commerce Application Adoption
- Effectiveness of AI-Based Educational Applications

Question 2: Questionnaire Validation

Using the questionnaire developed in Question 1:

1. Identify the variables being measured.
2. Classify each variable as **nominal, ordinal, interval, or ratio**, where applicable.
3. Explain the purpose of each section.
4. Conduct a small pilot survey.
5. Identify ambiguous, repetitive, or biased questions.
6. Revise the questionnaire based on the pilot feedback.

Section B: Research Proposal Using LaTeX

Question 3: Prepare a Research Proposal Using LaTeX

Select a research topic related to **Computer Applications** and prepare a complete research proposal using **LaTeX**.

The proposal should contain:

1. Title of the Research
2. Introduction
3. Background of the Study
4. Problem Statement
5. Research Objectives
6. Research Questions
7. Research Hypothesis, where applicable
8. Significance of the Study
9. Scope and Limitations
10. Literature Review
11. Research Methodology
12. Research Design
13. Population and Sample
14. Sampling Technique
15. Data Collection Methods

16. Data Analysis Techniques
17. Expected Outcomes
18. Work Plan/Timeline
19. References

Suggested topics:

- AI-Based Devanagari Numeric Symbol Recognition
- Smart Attendance System Using Facial Recognition
- Cloud Storage Usage among IT Students
- User Acceptance of AI-Based Educational Applications
- Cybersecurity Awareness among University Students
- E-Learning Application Adoption among University Students
- User Experience Evaluation of Mobile Banking Applications
- Machine Learning-Based Student Performance Prediction
- Sentiment Analysis of Student Feedback
- Usability Evaluation of University Websites

Section C: Literature Review and Citation Using Mendeley/Zotero

Question 4: Literature Review Using Mendeley or Zotero

Select a Computer Application research topic and collect **at least 10 relevant research articles** from reliable academic sources.

Using Mendeley or Zotero:

1. Create a research folder/library.
2. Import the selected research papers.
3. Organize papers using appropriate folders/tags.
4. Extract important information from each paper.
5. Identify research objectives, methodology, datasets, findings, and limitations.
6. Identify the research gap.
7. Insert in-text citations using the selected reference manager.
8. Generate the reference list automatically.
9. Apply an appropriate citation style such as **APA or IEEE**.
10. Prepare a literature review based on the collected papers.

Question 5: Citation and Referencing

Using Mendeley or Zotero, prepare a research document containing:

- At least 10 in-text citations
- At least 10 references
- Different types of sources such as journal articles, conference papers, and books
- Automatically generated bibliography

- Consistent citation style

Section D: Plagiarism Checking Using Turnitin or any other related tools

Question 6: Plagiarism Checking

Prepare a research report/proposal of at least **1500 words** on a Computer Application-related topic and check the document using **Turnitin**.

Perform the following steps:

1. Prepare the research document.
2. Submit/upload the document to the assigned Turnitin submission.
3. Generate the similarity report.
4. Examine the highlighted similarities.
5. Identify improperly cited or insufficiently paraphrased content.
6. Revise the relevant sections.
7. Add appropriate citations and references.
8. Resubmit the revised document where permitted.
9. Compare the original and revised similarity reports.
10. Submit the final report along with the similarity report.

Practical output:

Submit the research Proposal document, plagiarism/similarity report, and a brief explanation of the changes made after checking.

Section E: Statistical Data Analysis

E1. Descriptive Statistics

Question 7

For the following marks obtained by students in a programming examination:

10, 30, 50, 40, 20, 60, 70, 80, 90, 31, 25, 30

Calculate:

1. Mean
2. Median
3. Mode
4. First quartile (Q1/P25)
5. Second quartile (Q2/P50)
6. Third quartile (Q3/P75)

Perform the analysis using **Python/R/SPSS/Stata** and interpret the results.

E2. Chi-Square Test

Question 8: Chi-Square Test of Independence

A researcher wants to determine whether **gender and preference for an online learning application** are associated.

Gender	Prefer	Do Not Prefer	Total
Male	45	25	70
Female	35	15	50
Total	80	40	120

Using the **Chi-Square Test of Independence**:

1. State the null and alternative hypotheses.
2. Calculate expected frequencies.
3. Calculate the Chi-Square statistic.
4. Determine the degrees of freedom.
5. Determine the critical value/p-value.
6. Test the hypothesis at the 5% significance level.
7. Interpret the result in the context of Computer Applications.

E3. Correlation

Question 9: Correlation Analysis

The following data represent the **number of hours spent using an educational application (X)** and **student performance score (Y)**:

Student	X	Y
1	25	58
2	19	52
3	28	65
4	30	70
5	18	51
6	24	62

Calculate the Spearman's Rank Order Correlation between X and Y.

1. State the hypotheses.
2. Calculate the correlation coefficient.
3. Determine the direction and strength of the relationship.
4. Test the significance of the correlation.
5. Interpret the result in the context of educational applications.

E4. T-Test

Question 10: Independent Samples T-Test

A researcher wants to determine whether there is a significant difference in the **average programming test scores** of students taught using two different teaching methods.

Method A	Method B
65	72
70	75
68	70
75	78
72	80
69	74
71	77
73	76

Perform an **independent samples t-test**.

1. State H_0 and H_1 .
2. Calculate the mean and standard deviation of both groups.
3. Calculate the t-statistic.
4. Determine degrees of freedom.
5. Obtain the p-value.
6. Test at the 5% significance level.
7. Interpret the result.

Question 11: Paired Samples T-Test

The programming scores of 10 students were recorded **before and after using an interactive programming application**.

Perform a paired samples t-test to determine whether the application produced a statistically significant change in scores.

Include:

- H_0 and H_1
- Mean difference
- Standard deviation
- t-value
- Degrees of freedom
- p-value
- Decision
- Interpretation

E5. F-Test

Question 12: F-Test for Equality of Variances

Two groups of students use different programming environments. Their programming test scores are:

Group A: 72, 68, 75, 70, 74, 69, 76, 71

Group B: 65, 80, 70, 75, 68, 82, 73, 78

Perform an **F-test for equality of variances**.

1. State the hypotheses.
2. Calculate the variance of each group.
3. Calculate the F-statistic.
4. Determine the degrees of freedom.
5. Obtain the critical F-value/p-value.
6. Test the hypothesis at the 5% significance level.
7. Interpret the result.

E6. One-Way ANOVA

Question 13: One-Way ANOVA

A researcher wants to compare the effectiveness of **three different programming-learning applications**.

The test scores obtained by students are:

Application A	Application B	Application C
65	72	80
68	75	82
70	74	85
66	78	81
69	76	84

Perform a **one-way ANOVA**.

1. State H_0 and H_1 .
2. Calculate group means.
3. Calculate the grand mean.
4. Calculate between-group variation.
5. Calculate within-group variation.
6. Construct the ANOVA table.
7. Calculate the F-statistic.
8. Determine the p-value/critical value.
9. Test the hypothesis at the 5% significance level.

10. Interpret the result.

E7. Two-Way ANOVA

Question 14: Two-Way ANOVA

A researcher studies the effect of **programming language** and **teaching method** on student programming performance.

Prepare the appropriate **two-way ANOVA table** for the given experimental data.

Determine:

1. Whether programming language has a significant effect.
2. Whether teaching method has a significant effect.
3. Whether there is a significant interaction effect.
4. Appropriate hypotheses.
5. F-statistics and p-values.
6. Interpretation of the findings.